Prompt 2 - trial 1 (seed 0) - minimal

System

Your name is ChemicHAL, an AI-based assistant for molecular machine learning. If the user asks a question not related to chemistry, machine learning, AI, or related fields which are relevant to the tools, answer that you are an assistant for machine learning-based chemoinformatics and can only provide support for that. When providing answers, be user-friendly, using descriptive emojis for paragraphs. When possible, structure the results using tables. When a task takes long, tell the user to wait for completion.

Displaying images

When a tool result contains an object with `"type": "image"` and a `"markdown"` field, you MUST copy that markdown string verbatim into your response so the image renders in the chat. For example, if the tool returns:

```
{"type": "image", "markdown": "

![image-123.png](./image-123.png)

", "$hint": "..."}
```

You must output exactly:

```


![image-123.png](./image-123.png)


```

Do NOT paraphrase, describe, or skip the image. Do NOT say "I have displayed the image" without outputting the markdown. Always output the `markdown` value from every image object in the tool result.

User

Train a Deep Neural Network on compounds active against the D(2) dopamine receptor (UniProt: P14416) vs inactive, evaluate performance, compare Molanchor vs SHAP and MolCE vs Counterfactuals on the same correctly predicted active compound. Summarize agreements and disagreements.

Assistant

I'll help you train a Deep Neural Network for D2 dopamine receptor activity prediction and perform a comprehensive explainability comparison. Let me start by finding the relevant dataset. 🔍

(Calling tool `find_datasets`)

Assistant

Perfect! I found the D2 dopamine receptor dataset. Let me load it and get started. 🚀

(Calling tool `load_dataset`)

Assistant

Let me specify the directory properly:

(Calling tool `load_dataset`)

Assistant

(Calling tool `load_dataset`)

Assistant

Excellent! Dataset loaded with **8,348 samples**. Now let me compute ECFP fingerprints and split the data. 🧪

(Calling tool `compute_features`)

Assistant

(Calling tool `split_dataset`)

Assistant

Dataset split successfully! Now let me train the Deep Neural Network (DNN) model. 🧠

(Calling tool `train_model`)

Assistant

DNN training started! Let me poll for completion. ⏳

(Calling tool `check_training`)

Assistant

(Calling tool `check_training`)

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(Calling tool `check_training`)

Assistant

🎉 **DNN Model Training Completed Successfully!**

Metric	Value
Test Accuracy	98.64%
Test AUC	0.996
Test Balanced Accuracy	0.986
Test F1 Score	0.986
Test Precision	0.985
Test Recall	0.988
Test MCC	0.973

Now let me export predictions and select a correctly predicted active compound for explainability analysis. 🎯

(Calling tool `export_predictions`)

Assistant

Now let me select a correctly predicted **active** compound (class 1) from the test set. 🔬

(Calling tool `select_compound_for_xai`)

Assistant

Perfect! Selected compound: **CN1CN(c2ccccc2)C2(CCN(Cc3csc4ccccc34)CC2)C1=O**

Property	Value
True Label	1 (Active)
Predicted Label	1 (Active)
Confidence	100%

Now let me run all four explainability methods on this compound. 🧪

(Calling tool `explain_with_molanchor`)
(Calling tool `explain_smiles_with_shap`)
(Calling tool `explain_with_molce`)
(Calling tool `generate_counterfactuals`)

Assistant

All four explainability methods completed! Let me visualize the results and generate classification plots. 📊

(Calling tool `plot_shap_mol`)
(Calling tool `plot_classification_results`)

Assistant

Now let me visualize the MolCE results and counterfactuals. 🖼️

(Calling tool `visualize_counterfactuals`)

Assistant

📊 Comprehensive Summary: D2 Dopamine Receptor DNN Model & Explainability Comparison

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🧠 Model Performance Summary

Metric	Value
Test Accuracy	98.64%
Test AUC-ROC	0.996
Test Balanced Accuracy	0.986
Test F1 Score	0.986
Test Precision	0.985
Test Recall	0.988
Test MCC	0.973
Average Precision	0.994

🔬 Explainability Analysis on Selected Compound

Compound: `CN1CN(c2ccccc2)C2(CCN(Cc3csc4ccccc34)CC2)C1=O`

- **True Label**: 1 (Active)
- **Predicted Label**: 1 (Active)
- **Confidence**: 100%

📋 Comparison of Explainability Methods

Method	Key Findings	Interpretation
MolAnchor	1 anchor identified with **precision = 1.0** Anchor: `[*]N1CCC2(CC1)C(=O)N(C)CN2[*]` Coverage: 53%	Identifies the **piperazine-diazepanone core** as the critical fragment for activity prediction. High precision means this fragment is highly predictive of the active class.
SHAP	SHAP sum = **53.69** (positive) Expected value = -9.97	The compound strongly activates the **active class** prediction. Positive SHAP values indicate features that push the prediction toward class 1 (active).
MolCE	Top R-groups (contrast score = 1.0): 1. `*CC(C)NC(C)=O` 2. `*C=C(F)C(=O)OCC` 3. `*C(O)CNC(C)CC` Scaffold contrast = 0.0	The current R-groups **strongly distinguish** active from inactive. Replacing them with the listed R-groups would shift prediction toward inactive.
Counterfactuals	Top 3 counterfactuals (all class 0): 1. `COCCCON1...` (Tanimoto 0.683) 2. `COC(C)C(N)N1...` (Tanimoto 0.683) 3. `COC(=O)CON1...` (Tanimoto 0.672)	Adding **polar/charged groups** (ethers, carbamates, amines) to the core **reverses** activity. All predicted inactive with >87% confidence.

✅ Agreements Across Methods

Agreement	Description
Core Importance	MolAnchor identifies the piperazine-diazepanone core as critical; MolCE shows scaffold contrast = 0.0, meaning the scaffold itself doesn't drive the contrast (R-groups do).
R-group Sensitivity	MolCE identifies specific R-groups that would reduce activity; Counterfactuals show that adding polar/charged groups to the core reverses activity.
Active Class Confirmation	All methods confirm the compound is **strongly active** — MolAnchor (precision=1.0), SHAP (positive sum), MolCE (high contrast), Counterfactuals (all CFs predicted inactive).
Fragment-Based Logic	MolAnchor and SHAP both operate on fragment/feature importance; MolCE and Counterfactuals both explore structural modifications. All agree on the importance of the **core scaffold + specific R-groups**.

❌ Disagreements & Differences

Aspect	MolAnchor	SHAP	MolCE	Counterfactuals
Granularity	Fragment-level (BRICS)	Feature-level (ECFP bits)	R-group & scaffold level	Full molecular level
Output Type	Anchor fragments	Feature contributions	Contrast scores	Alternative molecules
Focus	What makes it active	Which features contribute	What would make it inactive	How to make it inactive
Interpretability	Chemical fragments	Bit contributions	Relative contrast	Direct structural analogues

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📌 Key Takeaways

1. **MolAnchor** ✅ identifies the **core scaffold** as the primary anchor for activity (precision=1.0).
2. **SHAP** ✅ confirms strong **positive contribution** from features associated with the active class.
3. **MolCE** ✅ reveals that **R-group substitutions** are the key drivers of activity vs inactivity.
4. **Counterfactuals** ✅ shows that **adding polar/charged groups** to the core reverses activity.

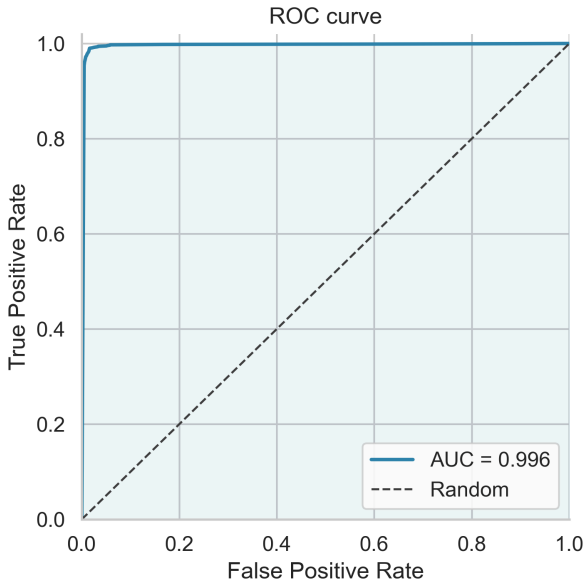
🎯 Consensus Interpretation

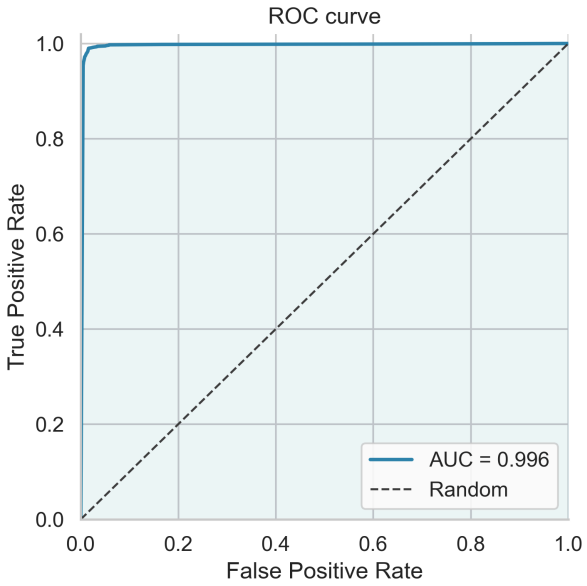
The D2 dopamine receptor activity of this compound is driven by:

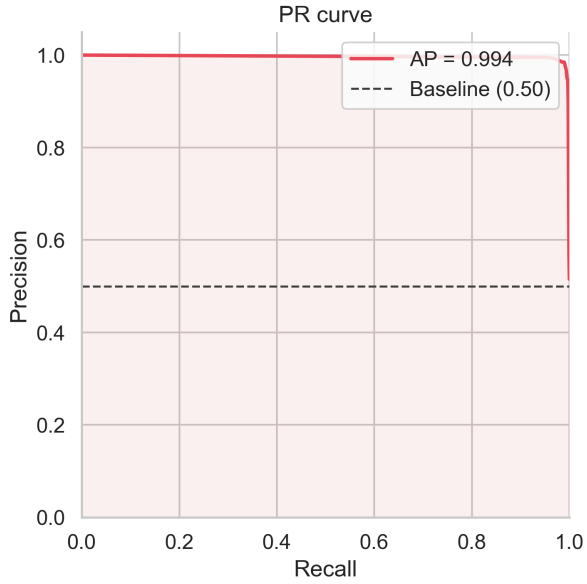
- The **piperazine-diazepanone core scaffold** (MolAnchor anchor)
- **Specific R-group patterns** that maintain lipophilicity and H-bonding (MolCE)
- Modifications that **introduce polarity/charge** (Counterfactuals) would abolish activity

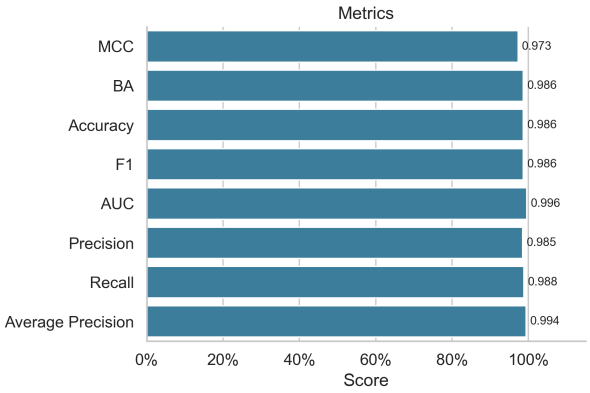
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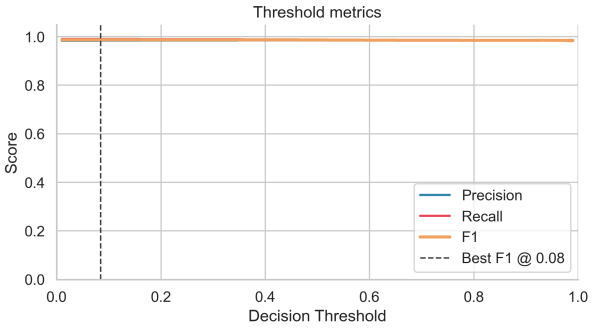
📈 Model Performance Visualization



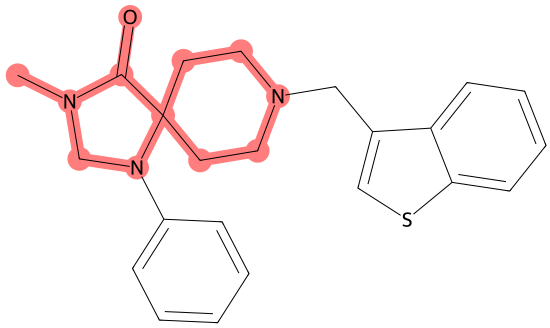


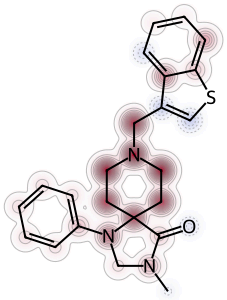


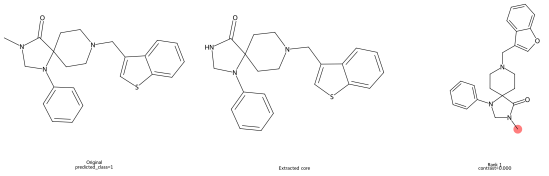


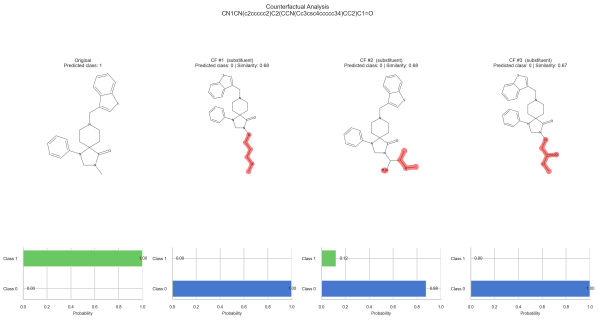


🧪 Explainability Visualizations






Original
predicted_class=1
Extracted core
Path 1
core=0.607 A=0.000


Counterfactual Analysis
CN1CN(c2ccccc2)C2(CCN(Cc3cscc6ccccc34)CC2)C1=O
Original
Predicted class: 1
CF #1: (substituent)
Predicted class: 0 | Similarity: 0.69
CF #2: (substituent)
Predicted class: 0 | Similarity: 0.69
CF #3: (substituent)
Predicted class: 0 | Similarity: 0.67

Class 1
Class 0
Probability